

SUSHMITHA HALLI SUDHAKARA

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PUBLICATIONS

- **S. H. Sudhakara**, M. S. Munir, M. Rahman and S. Shetty, "Native AI-based Predictive Operational Resiliency in Cyber-Physical Energy Systems," IEEE IWCMC 2025, Abu Dhabi, UAE, pp. 1258–1263. [Link](#)
- **S. Halli Sudhakara** and L. Haghnegahdar, "Security Enhancement in AAV Swarms: A Case Study Using Federated Learning and SHAP Analysis," IEEE Open Journal of Intelligent Transportation Systems, 2025. [Link](#)
- **Sudhakara, S. H.**, Haghnegahdar, L., GhasemiGol, M., Takabi, D., "Utilizing Federated Learning and SHAP for Predictive Analysis in Smart Grid Security," CSCE 2024, CCIS vol. 2251, Springer, Cham. [Link](#)

RESEARCH EXPERIENCE

Machine Learning Researcher, VMASC, Norfolk, VA

Jan 2024 – Jan 2025

- Implemented a Native AI framework for cyber-physical energy systems from a research direction defined by the principal investigator, built the full ensemble (Random Forest, XGBoost) and LSTM pipeline, achieving 99.97% fault-detection recall and reducing DER simulation downtime by 30%; accepted and presented at IEEE IWCMC 2025.
- Introduced Gini-Permutation feature selection and temporal encoding, improving cross-system generalization by 22%.
- Applied FedAvg and GANs to enhance smart-grid security on imbalanced datasets, achieving 81% accuracy, 85% F1, 78% precision, and 94% recall.
- Co-authored two research papers on federated learning and SHAP-based model interpretability for smart-grid and UA V-swarm security (CSCE'24; IEEE OJ-ITS).

Graduate Research Assistant, Old Dominion University, Norfolk, VA

Sep 2023 – Dec 2023

- **Built a labeled dataset for a menu-card segmentation study** - annotated menu-card regions across five menu-section categories in Labelbox, exported through the Labelbox SDK into COCO format, and assembled a Hugging Face datasets pipeline pairing images with category-labeled bounding boxes.
- Web-scraped 10,000+ Reddit entries with Python and BeautifulSoup to build a sentiment-analysis training dataset.

RESEARCH & APPLIED PROJECTS

Reproducibility Study: "Generating Long Sequences with Sparse Transformers" | [GitHub](#)

Apr 2024 – May 2024

Graduate deep learning course project (CS 795, ODU); owned the image-modality implementation

- **Reimplemented the fixed sparse attention factorization from Child et al. (2019) from the paper** for image data and trained on 4x NVIDIA A100s, reproducing the CIFAR-10 result to within 0.13 bits/dimension, 2.93 against the reported 2.80, on roughly 1/8 the training budget (15 epochs vs. 120).
- **Isolated training duration and model capacity, rather than the sparse attention mechanism itself, as the source of the replication gap**, by holding learning rate, dropout, layers, heads, and embedding size fixed to the paper's reported configuration and varying only what the available compute allowed - the original used 64 V100s over 7 days.

NYC Taxi Demand Forecaster — Serverless ML Pipeline | [GitHub](#)

Jan 2025 – Apr 2025

Hourly taxi-demand forecasting with a feature store and automated retraining

- **Reduced test MAE by 26%, from a 3.37 best baseline to 2.49**, by benchmarking naive baselines against XGBoost (2.60) and LightGBM (2.54), then tuning LightGBM with Optuna over a TimeSeriesSplit cross-validation objective to avoid look-ahead leakage.
- **Automated hourly feature ingestion and inference end to end** with two chained GitHub Actions workflows — a cron-scheduled feature pipeline writing to a Hopsworks online feature store, triggering an inference pipeline that loads the model from the registry and writes predictions back as a feature group.
- **Deployed a live Streamlit forecasting and monitoring app** backed by the feature store, joining stored predictions against observed ride counts to track model error over time; training and inference packaged as a Poetry-managed module with custom scikit-learn transformers.

AWS Serverless MLOps Pipeline — Vehicle Image Classification | [GitHub](#)

Sep 2024 – Oct 2024

AWS Machine Learning Engineer Nanodegree project

- Trained a bicycle-vs.-motorcycle image classifier with Amazon SageMaker's built-in image-classification algorithm (CNN); decoded 3,073-integer CIFAR-100 records into 32x32x3 RGB arrays, staged PNG images and TSV metadata in S3, and trained for 30 epochs on an ml.p3.2xlarge instance, reaching 81% validation accuracy.

- Deployed the model to a SageMaker endpoint and orchestrated inference with AWS Step Functions and three Lambda functions for data generation, classification, and confidence filtering; routed predictions below a 75% confidence threshold for review.
- Configured SageMaker DataCaptureConfig to capture 100% of inference requests and responses to S3, and built custom visualizations over the captured data to track model behavior.

INDUSTRY EXPERIENCE

Technical Sales Enablement Specialist (AI & Automation Projects), Decisions LLC

Jul 2024 – Present

- **Production prompt engineering:** Designed and iterated prompts extracting sales opportunities and classifying use cases from call transcripts; enforced JSON-only structured outputs with confidence scores and classification rationale, and debugged over-restrictive filtering to cut false negatives. Validated across GPT-4, GPT-5, and open-source models, version-tracked in GitHub.
- **LLM evaluation and failure analysis:** Investigated inconsistent GPT-5 ChatCompletion behavior (empty responses, truncation, tool-flow failures) across an internal AI assistant's tool flows; analyzed reasoning, tool-calling, and output-token behavior, then designed structured success/fallback JSON handling for empty, malformed, incomplete, and failed results, validated in QA.
- **Internal retrieval agent:** Built and shipped a conversational retrieval agent (Microsoft Copilot Studio + Power Automate) letting account executives query the internal demo library from Teams in natural language, replacing manual requests for industry- and use-case-specific materials.
- **Rule extraction automation:** Generated and tested rule sets from document extracts; iterated system prompts to improve extraction accuracy and output structure, and flagged missing inputs and undefined variables in specs back to stakeholders.
- **Feature validation and documentation:** Authored end-to-end documentation for AI platform features (agents, chat completion, embeddings, MCP servers) and the Deployment Server from hands-on testing of expected vs. actual behavior; partnered with Product, Engineering, QA, and Support across releases.

Data Science Intern, NightHack, Bengaluru, India

Jun 2023 – Aug 2023

- Applied PaddleOCR and Transformers for text extraction and tag classification across 2,000+ titles, scaling the solution to 10,000 titles; performed exploratory analysis and managed large-scale datasets in MongoDB.

Software Engineer — Cloud Services, LTIMindtree, Bengaluru, India

Sep 2019 – Jul 2022

- **Timecard web app:** Streamlined employee time tracking by building a Timecard web app integrated with Oracle Fusion Cloud (REST APIs), replacing a slow/error-prone workflow and delivering significant time savings and reduced labor costs by cutting time-to-submit, rework, and support effort. ([YT link](#))
- **Single Sign-On integration:** Implemented SSO for Oracle Fusion Cloud, removing separate credential management and reducing login friction across the user base.
- **Data pipeline and reporting:** Built custom HDL templates and fast formulas in Oracle SQL for bulk data loads, and developed custom reports against the Oracle Fusion database to support process decisions.

SKILLS

Deep Learning & Foundation Models: PyTorch, Hugging Face Transformers, sparse attention, multi-GPU training (4×A100), model interpretability (SHAP), CNNs, LSTMs, federated learning (FedAvg), GANs, ensemble methods

LLM & Agentic Systems: Prompt engineering, structured JSON outputs, embeddings, AI agents and tool use, MCP servers, RAG, vector retrieval (PostgreSQL/pgvector), GPT-4/GPT-5, Llama 3.2 (Groq API), LLM evaluation and failure analysis

ML Engineering & Cloud: AWS (SageMaker, S3, Lambda, Step Functions, EC2), GitHub Actions (CI/CD), Hopsworx feature store, Optuna, XGBoost, LightGBM, AutoGluon, scikit-learn, Streamlit, Poetry, Labelbox

Languages & Data: Python, SQL, NumPy, Pandas, PostgreSQL, Matplotlib, Plotly, Jupyter

EDUCATION

M.S. in Computer Science, Old Dominion University, GPA 3.88

Aug 2022 – Dec 2024

B.E. in Information Technology, Visvesvaraya Technological University, GPA 3.4

Jun 2015 – Aug 2019